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21 December 1984
SUPERSEDING
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MILITARY SPECIFICATION

FEATHERS AND DOWN, WATERFOWL; TAN-O-QUIL-QM TREATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers the processing and requirements for waterfowl feathers and down chemically modified by a TAN-O-QUIL-QM type treatment (see 6.3).

1.2 Classification. The waterfowl feathers and down shall be of the following types and classes as specified (see 6.2):

Type

I	-	Feathers
II	-	Down
III	-	Feather and down mixture

Class

1	-	Contractor furnished
2	-	Government furnished

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

V-T-276	-	Thread, Cotton
QQ-S-781	-	Strapping, Steel, and Seals
CCC-C-429	-	Cloth, Osnaburg, Cotton
CCC-C-432	-	Cloth, Sheeting, Cotton, (Unbleached, Bleached, and Dyed)
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-S-760	-	Strapping, Nonmetallic (and Connector)

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MIL-C-332	-	Cloth, Balloon, Cotton

STANDARDS

FEDERAL

FED-STD-148	-	Classification, Identification, and Testing of Feather Filling Material
FED-STD-191	-	Textile Test Methods
FED-STD-751	-	Stitches, Seams, and Stitchings

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MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 Laboratory report approval. Unless otherwise specified at the time of submission of bid (see 6.2), the bidder shall submit to the contracting officer a certified copy of a recent laboratory report covering the quality and grade of material which he proposes to deliver. The laboratory report shall contain test data which demonstrates that the finished product of the type which the contractor proposes to deliver has been tested in accordance with methods specified in 4.4.3 and found to comply with the requirements of this document. Any of the following types of reports will be satisfactory from the standpoint of this requirement:

(a) An independent or commercial laboratory report.

(b) The prospective contractor's own laboratory report.

(c) A Government laboratory report from a contract within six months of submission of bid.

The purpose of the above requirement is to assist the Government to determine the capability of the bidder to supply a product meeting all the requirements of this document. The submission of an acceptable report under this requirement shall not be construed as relieving the contractor from subsequently meeting all the requirements of this document.

3.2 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.4).

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3.3 Materials.

3.3.1 Class 1, contractor furnished materials. The class 1 materials shall be contractor furnished. The waterfowl feathers and down shall be new (not previously used or reclaimed). The feathers and down shall be of such composition and quality as to yield finished processed feathers and down meeting the requirements of this document.

3.3.1.1 Processing class 1 materials. The class 1 materials shall be processed in conformance with 3.4, 3.4.1 and 3.5.

3.3.2 Class 2, Government furnished materials (see 3.6.2.2). The class 2 materials shall be furnished by the Government. The types I and II Government materials shall be processed, sampled and packed in such a manner that the separate identity of each lot (GSA contract and "F" number within each group number) is preserved up to blending. The lot identification for type III Government materials shall be maintained as follows:

(a) The contractor shall submit detailed records of the components (ex: untreated and treated weights) in the respective blended lots. Each blended lot shall be ticketed and marked in accordance with MIL-STD-129.

(b) The contractor shall submit a certification of the weights of the treated feathers and down comprising each blended lot.

3.3.2.1 Processing class 2 materials.

3.3.2.1.1 Fractionation. The class 2 feathers shall be fractionated to remove quill feathers. The content of quill feathers in the final treated product shall not exceed 1.0 percent by weight when tested as specified in 4.4.3. Unless otherwise specified or approved by the contracting officer, the content of "usable" feather stock in the quill-waste fraction shall not exceed 12.0 percent by weight. For this purpose, "usable" feather stock shall be defined as specified in 4.4.2.1. The remaining processing for class 2 materials shall be in conformance with 3.4, 3.4.1 and 3.5.

3.4 Cleaning. The materials shall be dusted and washed with a detergent in water at approximately 100°F. The materials shall be washed and repeatedly rinsed until the rinse water is completely clean. The washed material shall have a turbidity value of not less than 75 centimeters when tested as specified in 4.4.2.2.

3.4.1 Drying (see 6.3.3). If the materials are to be subjected immediately to the chemical modification, they need not be dried. If the materials are dried prior to treatment, they shall not be dried to bone dry condition.

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3.5 Treatment. The washed product shall be given an approved TAN-O-QUIL-QM treatment (see 6.3). The use of treatment materials other than those approved is prohibited.

* 3.6 Finished product. The finished product shall conform to the applicable requirements in table I when tested as specified in 4.4.3.

TABLE I. Finished product requirements

	Type I		Type II		Type III
	Class 1	Class 2	Class 1	Class 2	
Oxygen number, max.	6.0	6.0	6.0	6.0	-
Brittleness	1/	-	-	-	-
Min. filling power in cm not laundered	-	-	7.5	4/	-
Min. filling power in cm after 2 launderings	6.0	4/	6.0	-	As in 3.6.2
Chrome content, percent 2/	0.45 ± 0.10	0.45 ± 0.10	0.40 ± 0.15	0.40 ± 0.15	-
pH	3.0 to 4.5	3.0 to 4.5	3.0 to 4.5	3.0 to 4.5	-
Odor 3/	None	None	None	None	-
Composition	As in 3.6.1.1	-	As in 3.6.1.2	-	-
Quill feathers, percent maximum	-	As in 3.3.2.1.1	-	-	-
Usable feather stock in quill waste fraction, percent max.	-	As in 3.3.2.1.1	-	-	-
Residual matter, percent max.	-	2.0	-	2.0	-
Blending	-	-	-	-	As in 3.6.2

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1/ The whole feathers shall be pliable and shall show no more than 8.0 percent failures (four fractured feathers).

2/ Expressed as chromic oxide.

3/ No odor of putrefaction.

4/ The filling power shall be determined and reported by the contractor, except when types I and II, class 2 are to be immediately blended by the contractor and furnished as type III material.

3.6.1 Composition.

3.6.1.1 Type I, class 1. The treated class 1 feathers shall conform to the following composition requirements when tested as specified in 4.4.3. Classification of components shall be in accordance with the applicable definitions in FED-STD-148.

(a)	Whole waterfowl feathers and pure down (and plumules)	Minimum 90.0 percent
(b)	Pure down (and plumules)	Minimum 10.0 percent
(c)	Balance composed of any combination of the following:	Maximum 10.0 percent
	1. Non waterfowl feathers	Maximum 8.0 percent
	2. Waterfowl quill feathers	Maximum 1.0 percent
	3. Feather fibers, or damaged waterfowl feathers	Maximum 5.0 percent
	4. Down fiber	Maximum 10.0 percent of the down content
	5. Residual matter	Maximum 2.0 percent

3.6.1.1.1 Waterfowl feathers (whole). A minimum of 50 percent by weight of all feathers (type I, class 1) shall measure less than 3 inches in length. Feathers longer than 4 inches but not exceeding 6 inches shall total not more than 5 percent. The balance shall be composed of feathers not exceeding 4 inches in length.

3.6.1.2 Type II, class 1. The treated class 1 down shall conform to the following composition requirements when tested as specified in 4.4.3. Classification of components shall be in accordance with the applicable definitions in FED-STD-148.

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(a)	Pure down (and plumules)	80.0 percent, minimum
(b)	Extraneous (total of components below)	20.0 percent, maximum
	1. Small waterfowl feathers 1/	15.0 percent, maximum
	2. Down fiber	8.5 percent, maximum
	3. Feather fiber	5.0 percent, maximum
	4. Quill and damaged feathers, non-waterfowl feathers	1.0 percent, maximum
	5. Residual matter	2.0 percent, maximum

1/ Feathers over 2.5 inches not permitted.

3.6.2 Type III blending.

3.6.2.1 Class 1. A blend of chemically modified waterfowl feathers conforming to type I, class 1 and waterfowl down conforming to type II, class 1 shall be prepared by blending not less than 10 percent of waterfowl down (type II, class 1) with the waterfowl feathers (type I, class 1), so that the resulting blend shall have a filling power of 6.0 ± 0.2 cm, after two launderings when tested as specified in 4.4.3 (see 6.3.3).

3.6.2.1.1 Maximum percentage of down. Under no circumstances shall the total percentage of down (type II, class 1) in blend exceed 40 percent.

3.6.2.2 Class 2. Chemically modified waterfowl feathers conforming to type I, class 2 shall be blended with chemically modified waterfowl down conforming to type II, class 2. The percentage by weight of feathers in the blend shall not exceed 50 ± 5 percent. Unless otherwise specified, the lots of Government furnished feathers and down (see 3.3.2) shall be blended in accordance with blending list, in groups by lots, as furnished by the contracting officer. The contractor shall perform filling power tests after laundering on the final 50/50 blended lots as specified in 4.4.3 and record results in addition to the identification records required in 3.3.2. Filling power data for the final 50/50 blend shall be furnished to the Government for information purposes only.

3.7 Workmanship. The finished product shall be uniformly processed, treated, of uniform composition, and shall conform to the quality of product established by this document. The occurrence of defects shall not exceed the applicable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government

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reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for compliance with 4.4.2, and tested for the characteristics specified in table III. The presence of non-compliance with the in-process inspection or failure of any test shall be cause for rejection of the first article.

* 4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document. In addition, the woven polypropylene bag material shall be tested in accordance with the applicable test methods of FED-STD-191 as indicated in table II. All test reports shall contain the individual values used in expressing the final result. The sample unit shall be 1/2 yard full width for the polypropylene fabric. The lot size shall be expressed in units of 1 yard and the sample size shall be as specified below. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified.

Lot size (yards)	Sample size (number of sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE II. Component tests

Characteristic	Requirement paragraph	Test method
Polypropylene fabric:		
Material identification	5.1.1.2.1	1/
Weave	5.1.1.2.1	Visual
Weight	5.1.1.2.1	5041
Yarns per inch	5.1.1.2.1	5050
Breaking strength	5.1.1.2.1	5100
Seam efficiency	5.1.1.2.1	5110
Air permeability	5.1.1.2.1	5450
Ultraviolet exposure	5.1.1.2.1	5804
Color	5.1.1.2.1	1/
Polypropylene yarn (thread):		
Yarn identification	5.1.1.2.1.1	1/
Denier	5.1.1.2.1.1	1/
Breaking strength	5.1.1.2.1.1	1/
Color	5.1.1.2.1.1	1/

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

4.4.2 In-process inspection. Inspection shall be conducted to see that the processing is in accordance with the specified requirements. The Government reserves the right to exclude from consideration for acceptance any material or services for which in-process inspection has indicated nonconformance. The contractor shall certify conformance to the following requirements:

(a) Use of new feathers, new down or new feathers and down (see 3.3.1).

(b) Use of approved chemical compounds, in required ratios (see 3.5).

(c) Other processing details required in this document.

4.4.2.1 Usable stock in quill waste (type I, class 2). Quill waste shall be examined for content of "usable" feather stock (see 3.3.2.1.1). The quill waste accumulated from each lot (GSA contract and "F" number) shall be considered a lot for this examination. A composite minimum 24-ounce sample shall be made up of samplings of not less than 4 ounces each from various portions

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of the lot, and thoroughly mixed. From this sample, a specimen of not less than 8 grams shall be taken and weighed to the nearest 0.01 gram. The usable stock consisting primarily of feathers (excluding quill feathers) and down (and plumules) shall be separated from the specimen. For the purpose of this determination "usable" stock components are characterized by a slower rate of fall, compared to obvious quill feathers, when dropped from a height of 18 inches or more. Tangles and clusters which are not easily separated for individual component classification, shall be classified as "usable" stock, or as quill feathers according to the rate of fall (these distinctions are in recognition of the nature and capabilities of the fractionation process). The "usable" stock components shall be weighed to the nearest 0.01 gram, and the content calculated and reported to the nearest 0.1 percent based on the original specimen weight. Except as otherwise specified or accepted by the contracting officer, quill waste which contains more than 12.0 percent of "usable" stock shall be refractionated to recover the excess.

4.4.2.2 Turbidity testing of cleaned feathers and down. The cleaned feathers and down (before chemical modification treatment) shall be tested for turbidity value in accordance with Method 10 of FED-STD-148. The sample shall be a 20-ounce minimum composite sample made up of samplings from the several washer loads in each 1,000 pounds of product or fraction thereof which shall be considered a lot for purposes of the turbidity test. The composite sample shall be dried, conditioned, and mixed with a rod to insure adequate composite, before taking the specimen for test. It is not required under quality assurance procedures that the turbidity test be completed prior to commencement of the chemical modification treatment. However, failure of the lot to meet the 75 centimeter turbidity value requirement shall be cause for rejection of the finished product corresponding to the in-process testing lot, regardless of conformance to specified finished requirements.

4.4.3 End item testing. Testing of the finished product shall be performed in accordance with the methods specified in table III. The physical and chemical values specified in section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values used in expressing the final result. The test methods indicated by number shall be performed as specified in FED-STD-148. The sample unit for testing shall be a throughly mixed 24-ounce composite of the finished product. Not more than 6 ounces shall be taken from any level A (see 5.1.1) unit pack (bag). For larger bags, sampling shall be from top, center, bottom, and opposite sides of the bag, with not more than 12 ounces from any one bag. Care shall be taken in mixing to prevent damaging or compressing the material. The sample size and the acceptance number shall be in accordance with the following:

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Lot size (pounds)	Sample size (sample units)	Maximum acceptable number of test failures
3,200 or less	2	0
3,201 up to and including 10,000	3	0
10,001 up to and including 35,000	5	0
35,001 and over	8	1

TABLE III. End item tests 1/

Characteristic	Requirement paragraph	Test method
Quill feathers	3.3.2.1.1	2
Usable feather stock in quill waste fraction	3.3.2.1.1	4.4.2.1
Residual matter	3.6	2
Oxygen number	3.6	4
Brittleness	3.6	3
Filling power (not after laundering)	3.6	1
Filling power, after laundering	3.6	4.4.3.1
Chrome content	3.6	9
pH	3.6	8
Odor	3.6	11
Composition	3.6	2

1/ Testing may be performed under prevailing conditions except that in settlement of dispute cases tests shall be conducted under standard conditions as defined in FED-STD-148.

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4.4.3.1 Laundering procedure. Place approximately 1 ounce of the composite sample in a 17 by 6 inch cloth bag made of water repellent and preshrunk cloth conforming to type I, class 2 of MIL-C-332. The bag shall be tightly seamed and closed to prevent loss of material during laundering. The filled bag shall then be subjected to two successive complete launderings (wool method) in accordance with the general procedure of Method 5556 of FED-STD-191. The dryer shall be adjusted to have a stack temperature of 150° to 170°F. After the two wash cycles, the material shall be removed from the bag, conditioned, and tested for filling power in accordance with Method 1 of FED-STD-148.

4.4.4 Packaging inspection.

* 4.4.4.1 Inspection of fabricated bags and baling boards. Fabricated bags and baling boards shall be inspected to determine conformance with the applicable requirements of 5.1 and 5.2. Defects shall be scored in accordance with the list below. The lot size shall be expressed as fabricated bags or baling boards. The sample unit shall be one bag or baling board. The inspection level shall be S-2, and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0.

Examine	Defect
Bag construction:	Finished length and width not as specified. Seam or stitch type not as specified.
Cloth bag	Seams on sides and bottom have less than 5 or more than 6 stitches per inch. Margin of stitching not as specified. Gage of stitching not as specified. End of stitching when not caught in another seam or stitching, not finished as specified. Thread type and ply not as specified. Thread breaks not backtacked or overstitched as specified. Loose tension resulting in a loose seam or stitching. Thread ends not trimmed.
Polypropylene bag	Bag not one piece construction. Seams on two long sides have less than 4 or more than 6 stitches per inch. Raw edges not turned in and stitched as specified. Ends of stitches not secured at beginning or end by a 1 to 1-1/4 inch long chain off the fabric. Open top end of bag not finished as specified. When specified, bags not turned so that seams are on the inside of bag.

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Examine	Defect
Baling boards	Baling board material not as specified. Size of baling boards not as specified. Strapping material not as specified.

4.4.4.2 Weight examination. The lot shall be unacceptable if the content of the bags in the sample is less than the total content weights indicated on the bags. The lot size shall be expressed in units of bags. The sample unit shall be one filled bag. The number of bags to be selected for the weight examination shall be in accordance with inspection level S-4. When specified, or in case of dispute, the weight basis for acceptance shall be determined by correction of unconditioned weight, to weight at equilibrium under standard atmospheric conditions as defined in FED-STD-148. At the time of weighing, a random sample of product not less than 1/4 pound shall be taken from the contents of the bag, and placed immediately in a tared closed container. The sample shall then be weighed in the closed container. The sample shall then be spread in a shallow layer in a tray and stored in the standard atmosphere for not less than 24 hours with occasional stirring. The conditioned sample shall then be weighed and the gain or loss calculated to a percentage based on the original sample weight. The percentage correction shall then be applied to the unconditioned weight of the contents of the bag to determine its weight on the standard weight basis.

4.4.4.3 Preservation examination. The bags of feathers shall be examined for the defects listed below. The lot size shall be expressed in units of bags. The sample unit shall be one bag of product. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Defects
Not preserved in accordance with specified preservation level.
Bag not new, or any patch.
Filled bag not closed as specified.
Wire ties not as specified.
Application of ties to bag not positioned as specified.
Wire tie broken or not tightly turned to accomplish a secure closure.
Any hole, rip, tear, open seam; any weak places in bag which may develop into a hole; backtacking and backstitching not as specified.
Bag wet.

4.4.4.4 Packing examination. The bales or shipping containers shall be examined for the defects listed below. The lot size shall be expressed in units of bales or shipping containers. The sample unit shall be one bale or shipping container. The inspection level shall be level I and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

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Defects

Not packed in accordance with specified packing level.
Bales not compressed to specified size.
Does not contain the specified number of bags.
Kraft facing of baling board not on outside.
Strapping not affixed girthwise.
Less than eight straps used on each bale.
The two outside straps on each bale not located as specified from each end.
The six intermediate straps not placed equidistant between the two outside straps.
Any board or strap too loose or not placed to assure safe delivery of contents
Marking omitted, incomplete, illegible, incorrect, not as specified, or not in proper location.

5. PACKAGING

* 5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

* 5.1.1 Level A. Thirty-five ± 5 pounds of type I or III product or twenty-five ± 3 pounds of type II product shall be unit packed in a cloth or polypropylene bag as hereinafter specified.

5.1.1.1 Bag, cloth.

5.1.1.1.1 Bag, material. The cloth bag shall be a new (not used or laundered) plain textile bag fabricated from cotton sheeting conforming to type VIII, class 1 of CCC-C-432 or osnaburg conforming to type I, class 2 of CCC-C-429. Fabrics of equivalent weight and construction will be acceptable.

* 5.1.1.1.2 Bag, construction. The bags shall have a finished length and width measured flat, of approximately 72 by 36 inches and shall be constructed in accordance with the following requirements: The side and bottom shall be seamed with 5 to 6 stitches per inch in seam type SSa-2, with either stitching type 401 or 301, conforming to FED-STD-751. The stitching shall be at least 5/8 inch from the cut edge of the fabric or 1/4 inch from a selvage edge and the rows shall be at least 1/8 inch apart. The ends of all stitching, when not caught in other seams or stitching, shall be backstitched or backtacked at least 1 inch. The thread used for seaming shall be cotton, machine type IA1 or IB1, unbleached, ticket No. 12, 4 ply conforming to V-T-276.

* 5.1.1.1.2.1 Type 301 stitching. Thread breaks in stitching shall be backstitched not less than 1 inch at each break. Thread tension shall be maintained so that there will be no loose stitching. Thread ends shall be trimmed.

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* 5.1.1.1.2.2 Type 401 stitching. Thread breaks in stitching shall be overstitched not less than 1-1/2 inches at each break. Thread tension shall be maintained so that there will be no loose stitching. Thread ends shall be trimmed.

5.1.1.2 Bag, woven polypropylene.

* 5.1.1.2.1 Bag material. The polypropylene bag shall be fabricated from new first quality materials. No scrap, reground or reworked polymer shall be used. The fabric shall be of woven yarns, filaments, ribbons or combinations thereof and be buff or natural in color. The yarns for both warp and filling of the fabric shall be made from a 100 percent polypropylene base resin containing an inhibitor for resistance to ultraviolet light. The finished fabric shall conform to the physical requirements specified in table IV when tested as specified in 4.4.1. The polypropylene fabric shall retain not less than 70 percent of its original tensile breaking strength in each direction after 100 hours ultraviolet exposure in a weatherometer. The finished fabric shall have a seam efficiency of not less than 60 percent.

TABLE IV. Physical requirements of the polypropylene bag material

Weight oz per sq yd minimum	Yarns per inch minimum		Breaking strength pounds minimum		Air permeability cu ft/min/sq ft maximum	Weave
	Warp	Filling	Warp	Filling		
2.4	12	8	115	80	27	Plain

5.1.1.2.1.1 Thread. The thread used for all stitching and seaming of the polypropylene bag shall be a nominal 1000 denier polypropylene yarn with a breaking strength of 5.0 grams per denier and shall be natural or white in color.

5.1.1.2.2 Bag construction. The bags shall have a finished length and width measured flat, of approximately 72 by 36 inches and shall be constructed in accordance with the following requirements: The bags shall be of one piece construction, closed on the two sides and bottom by folding the fabric at the bottom end and seaming the two long sides. the sides shall be seamed with 4 to 6 stitches per inch in seam type SSn-1, stitch type 401 conforming to FED-STD-751. The raw edges shall be turned over 1 to 1-1/4 inches and stitched with a row of stitching no nearer than 1/2 to 5/8 inch from the turned edge. Ends of stitches shall be secured either by beginning or ending in a chain 1 to 1-1/4 inches long off the fabric. The open top end shall be finished with a hem of 1 to 1-1/4 inches wide, seamed with 1 row of stitch type 401 with 4 to 6 stitches per inch, 1/2 to 5/8 inch from the turned edge. The ends of the stitches shall be secured by running 1 to 1-1/2 inches of chain off the fabric, or the last end of stitches may be stitched over the first end for a distance of 1 to 2 inches. Unless otherwise specified (see

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6.2), the bags shall be turned so that the seams are on the inside. The breaking strength of any seam shall not be less than 60 percent of the breaking strength of the fabric.

5.1.1.3 Closure. The closure of the top of the bag after filling shall be effected by means of two double loops of 0.072 inch thick galvanized soft iron or steel wire ties of not less than 6 inches long, having a 1/2 inch diameter eye formed at each end. One tie shall be applied at the base of an ear formed by gathering the cloth evenly and the second tie applied at a distance approximately 1 inch from the first tie with the twisted ends of two ties alternated and facing in opposite direction.

* 5.1.2 Commercial. Feathers and down shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be levels A, B or Commercial as specified (see 6.2).

5.2.1 Levels A and B packing. Three bags of type I or III product or four bags of type II product (see 5.1.1) shall be placed between two baling boards and compressed by machine so as to form a bale approximately one-fifth of the original volume. Bales thus formed shall measure approximately 60 by 24 by 20 inches.

5.2.1.1 Baling boards. Baling boards shall be solid fiberboard conforming to type SF, class weather-resistant, grade V4s of PPP-F-320. Two such baling boards, size 58 by 36 inches, each with a score line placed 6 inches from each 58-inch edge, shall be used in each bale; one board on top of bags being baled, and one on the bottom. The boards shall be bent at the score lines thereby forming a fiberboard cap 58 inches in length, 24 inches in width, with 6-inch flanges extended down over the side of the bale.

5.2.1.2 Strapping. Not less than eight straps shall be used on each bale; straps shall be affixed girthwise around the sides, top, and bottom of the bale over the baling boards. The two outside straps shall be placed 4 inches in from each end and the intermediate straps placed equidistant between the end bands. All strapping shall be flat not less than 3/8 inch wide by 0.020 inches thick conforming to class 1, type I, finish A of QQ-S-781 or type II of PPP-S-760.

* 5.2.2 Commercial packing. Feathers and down, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

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* 5.3 Marking. In addition to any special marking required by the contract or purchase order, interior packs and shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable. Each interior pack shall have attached a manila colored paper shipping tag for the identification information.

6. NOTES

6.1 Intended use. The treated feathers and down are intended primarily as filling material for sleeping bags, pillows and other items. The type II product is intended specifically as filling material for sleeping bags.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this document.
b. Type and class of product required (see 1.2).
c. When laboratory report approval is not required (see 3.1).
d. When a first article is required (see 3.2, 4.3 and 6.4).
e. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
f. When bag seams are not required to be on the inside (see 5.1.1.2.2).

6.3 Approved "TAN-O-QUIL-QM" treatment. The "TAN-O-QUIL-QM" type treatment consists of the co-application of chrome "tanning" and water-repellent compounds under controlled conditions to enhance the effective bulk of the feathers and to improve their hygienic properties. Approval of such compounds and their applications for use under this document is the responsibility of the U.S. Army Natick Research and Development Center, Natick, MA 01760-5014 and is based on extensive tests, including those for toxicity, which are not set forth in this document. Because of the time necessary to conduct full evaluation, only those chemicals already approved for the standard "TAN-O-QUIL-QM" treatment and so listed in the invitation for bid or request for proposal shall be considered acceptable for the related product.

6.3.1 Materials. The standard approved "TAN-O-QUIL-QM" treatment makes use of a basic chrome sulfate as the "tanning" agent. The repellent is a myristylato chromium complex normally supplied as 30 percent solids in isopropyl alcohol. Common salt and sufluric acid are also required.

6.3.1.1 Preparation of materials. The approved chrome "tanning" agent is predissolved in boiling water, at approximately 10 pounds per 25 gallons. The approved water-repellent is predissolved in boiling water, at approximately 3 pounds per 25 gallons, by adding under the surface of the water through a tube and continuing boiling for at least 5 minutes after the addition has been completed.

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Precaution: Isopropyl alcohol will be volatilized during this preparation. The working area should be adequately ventilated, and the presence of open flames or sparks (e.g., open-type electric motors) should be avoided.

The repellent should be used within 1/2 hour of preparation.

6.3.2 Procedure. Immerse the washed feathers or down in water at 100°F using a ratio of three to four gallons of water to one pound of waterfowl feathers or down. The weight of the feathers and down shall be calculated on a dry basis. Add 20 pounds of salt for every 100 pounds of feathers or down and sufficient sulfuric acid to bring the bath to a pH of 3.4 to 3.6. Add 10 pounds (as supplied basis) of "tanning" agent (approved) and 3 pounds (as supplied basis) of commercial repellent (approved) per 100 pounds of feathers or 10 pounds (as supplied basis) of "tanning" agent (approved) and 3/4 pound (as supplied basis) of commercial repellent (approved) per 100 pounds of down. Hold with intermittent agitation until the turbidity of the bath has cleared up. This usually occurs within one hour and is considered to have been reached when newsprint can be read through a clear glass container, containing a sample of the bath (approximately 3 inches high) placed directly on the print to be read. Rinse feathers or down, drain, extract and dry. Feathers should be removed from the drier as soon as they are dry. Care should be exercised so as not to over-dry them.

6.3.3 Special notes.

(a) Thorough cleaning of the feathers and down prior to treatment is essential to achieving full benefit from the treatment. The intermediate turbidity test requirement (see 3.4) is a means of assurance that the effect of the treatment will not be materially lowered by the presence of extraneous material.

(b) Conformance to the materials and procedures described above does not relieve contractors from conformance to other requirements of this document. Characteristics of processing equipment and conditions, as well as the characteristics of the unprocessed feathers, must be considered in controlling the processing to assure satisfactory results.

(c) A proper balance between down quality, down content and waterfowl feathers should be achieved in order to maintain the required filling power of 6.0 $\pm$ 0.2 cms. Excessive filling power, which may appear to be desirable, results in bags whose resilliency and bulk are too lofty. This renders the bags difficult to roll and pack for transport.

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6.4 First article. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should consist of a 1-pound sample of finished product. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for inspection and approval of the first article.

6.5 Changes from previous issue. The margins of this document are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - NU
Air Force - 99

Preparing activity:

Army - GL

Project No. 8320-0086

Review activities:

Army - MD
Navy - MC
Air Force - 82
DLA - CT

User activity:

Navy - MS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-43097H	Feathers & Down, Waterfowl; Tan-O-Quil-QM Treated

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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